Platelet transfusion

Use in conjunction with local guidance and expert advice

- Considerations include symptoms, signs, laboratory parameters, bleeding history, medications, patients' values and preferences, alternative therapies, and overall clinical context.[1]
- Assessment of bleeding risk is complex and requires careful consideration. Specialist haematology advice may be required.[2]
- For neonates and paediatric patients, refer to specific guidelines and expert advice.
- Platelet transfusions are not indicated in all causes of thrombocytopenia and may be contraindicated in certain conditions (e.g. TTP and HIT).[2] The cause of thrombocytopenia should be established and expert opinion sought.[2]
- Seek expert guidance on indications and risk of platelet transfusion in the presence of platelet dysfunction or antiplatelet agents.[1,2]

Usual adult dose is 1 unit (pack) which should increase the platelet count by 20 – 40 x 10^9/L.

Platelet transfusion indications in adults

Prophylactic use to prevent bleeding No bleeding or minor bleeding such as skin petechiae (or WHO grade 1)	**Platelet count threshold x 10^9/L**
Reversible or chronic bone marrow failure (disease or therapy-related eg. chemotherapy/haematopoietic SCT)^	
• without additional risk factors for bleeding	10
• with additional risk factors e.g. with fever/infection, minor bleeding (per local guidance)	10 – 20
Critically ill patients depending on additional risk factors for bleeding (per local guidance)	$10^{(3)}$ – 20
Patients at high-risk of spontaneous bleeding (e.g. recent major or critical site bleeding or surgery, additional haemostatic defect/s or anticoagulation)	Count variable per expert guidance

^A strategy of no prophylaxis may be appropriate for selected patients with hypoproliferative thrombocytopenia,[1,4] management should be individualised in consultation with a relevant expert.[2]

Therapeutic use to treat active bleeding Clinically significant bleeding (e.g. WHO grade 2 or above)	**Platelet count threshold x 10^9/L**
⚠ In Major haemorrhage, refer to local Major Haemorrhage Protocol (MHP)	Directed by local MHP
Bleeding (non-neuraxial) — threshold individualised based on nature, severity and clinical context	30 – 50
Bleeding associated with critical sites (e.g. central nervous system)*	100

*No platelet transfusion is recommended in adults with spontaneous or traumatic, nonoperative ICH with platelet counts >100 x 10^9/L including those receiving antiplatelet agents.[1] Refer to local expert guidance.

Pre-procedure, individualised with expert local guidance# to prevent bleeding expected to occur during surgery/invasive procedure	**Platelet count threshold x 10^9/L**
Central Venous Catheter (CVC) placement at anatomical sites amenable to manual compression[1]	10
Other low risk CVC insertion; low risk procedures	20
Lumbar puncture[1,4]	20 – 40
Major surgery (non-neuraxial); vaginal or caesarean birth**; high risk procedures	50
High risk surgery at critical sites (e.g. neurosurgery or some ophthalmic surgery)	100

#Threshold individualised based on additional clinical features, bleeding risk, nature of procedure with expert advice or local guidance.
**Generally considered acceptable however lower counts may be tolerated. Seek expert advice.[2] For neuraxial anaesthesia, refer to local expert guidance.

References

1 Metcalf et al. Platelet Transfusion: 2025 AABB and ICTMG International Clinical Practice Guidelines JAMA. May 2025.

2 NBA Patient Blood Management Guidelines: Perioperative (2012), Medical (2012), Critical Care (2012), Obstetrics & Maternity (2015).

3 Coz Yataco A et al. Transfusion of Fresh Frozen Plasma and Platelets in Critically Ill Adults: An American College of Chest Physicians Clinical Practice Guideline. Chest. 2025.

4 Estcourt et al. Guidelines for the use of platelet transfusions, British Society of Haematology. Br J Haematol. Feb 2017.